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Regularization

Machine Learning Lecture Slides

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Regularization

1. Want to **remove** features that are not important.
2. Want to choose models that **generalise** better to unseen data.
3. Want to **bias** towards (prefer) certain types of models → ***model complexity***
4. **Regularization** provides a mechanism to address these.

Shrinkage functions

- One way to impose a preference over simpler models is to prefer models that have sparser/smaller parameter values

Use only λ

- A general form for such models can be expressed as:

$$\boldsymbol{L'(\theta) = \lambda L(\theta) + (1 - \lambda)\sum_j \left|\theta_j\right|^p}$$

where **λ** **(0 ≤ λ ≤ 1)** is a strength parameter and

$\sum_j \left|\boldsymbol{\theta}_j\right|^{\boldsymbol{p}}$ is the **Regularization** parameter

- **λ** controls the **trade-off** between **simpler models** vs. **the fit** with respect to the data.
- **λ = 0 vs. λ = 1**

Choosing λ

$$L(\theta) = \lambda\left(\sum_i loss\big(y_i, f(x_i)\big)\right) + (1-\lambda)\sum_j |\theta_j|^p$$

- **λ = 0 :** ignore fit choose θ = 0
- **λ = 1 :** choose best model according to ***loss***
- **λ = 0.5:** choose model that gives equal weight to smaller model and fit
- In practice, we need to find **λ** using cross-validation.
- Regularization has the effect of shrinking the parameters. Hence, also known has *shrinkage*.

3 Different Regularization Methods

$$\boldsymbol{L'(\theta) = \lambda L(\theta) + (1 - \lambda)\sum_j |\theta_j|^p}$$

- **p=0** : *Subset selection*
- **p=1** : *Lasso*
- **p=2** : *Ridge*

In each case, **λ** needs to be either:

- assumed i.e. given or
- chosen experimentally

p = 0 : Subset selection

- For p = 0: simply counts the number of non-zero features

$$L(\boldsymbol{\theta}) = \boldsymbol{\lambda}\left(\sum_{i} \boldsymbol{loss}\left(\boldsymbol{y_i}, \boldsymbol{f}(\boldsymbol{x_i})\right)\right) + (\mathbf{1} - \boldsymbol{\lambda}) \sum_{j} \left|\boldsymbol{\theta_j}\right|^{\mathbf{0}}$$

- Prefer models with less number of features
- i.e. prefer 'simpler' models with ***lesser*** number of parameters
- But function is not convex

Note: Here $\mathbf{0}^{\mathbf{0}} = \mathbf{0}$ while $\boldsymbol{x}^{\mathbf{0}} = \mathbf{1}$ for $x > 0$

Greedy Subset selection

- Input **(X,Y)**, and **K = no. of features to select**
- Initially, all parameters are set to 0 i.e. $\theta = 0$
- Repeat until K features chosen:
  - Select feature F that gives the greatest reduction in error
  - Apply 1 gradient descent step along F (while keeping other features unmodified)
- When K features have been selected:
  - apply gradient steps only

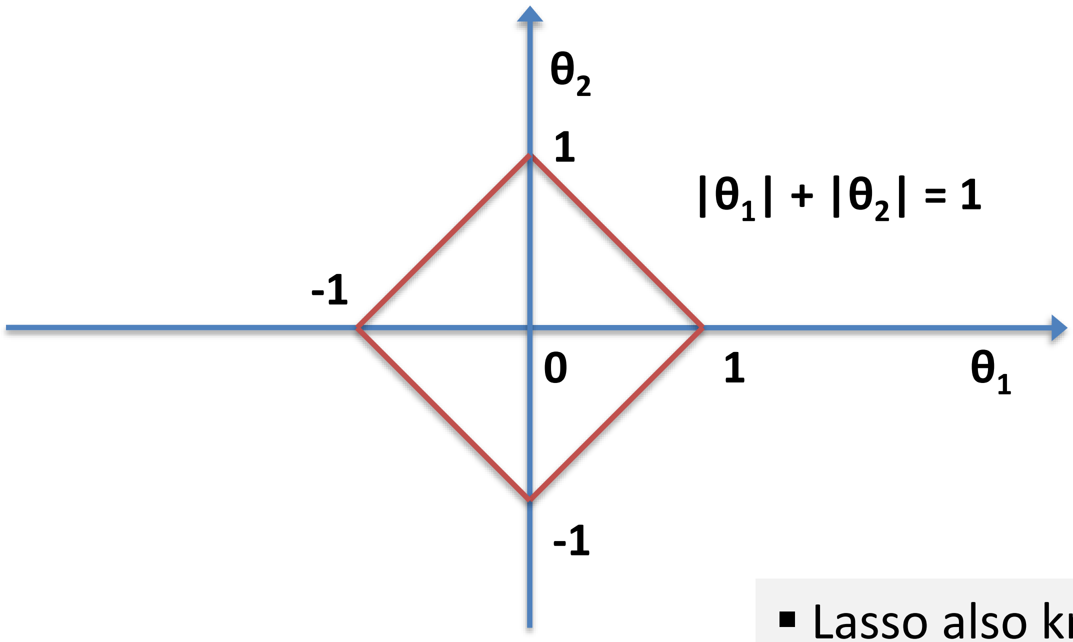
- Approximation for **p=0** case
- Requires deciding **K** in advance.
- That is not necessarily a disadvantage. Can find K using other methods – cross validation.
- Greedy method – hence can get stuck in locally optimal K

Possible question for open asessment

p = 1 : LASSO Regularized Regression

$$L(\theta) = \lambda\left(\sum_i loss(y_i, f(x_i))\right) + (1-\lambda)\sum_j |\theta_j|$$

- Smooth penalty over values of θ_j



Unit diamond

In general:

equilateral polytope

- Lasso also known as L1 Regularization

Computing gradient

$$L(\theta) = \lambda\left(\sum_i loss(y_i, f(x_i))\right) + (1-\lambda)\sum_j |\theta_j|$$

$$|\theta| = \left(\theta^2\right)^{\frac{1}{2}}$$

$$\frac{d}{d\theta}|\theta| = \frac{d}{d\theta}\left(\theta^2\right)^{\frac{1}{2}} = \frac{1}{2}\left(\theta^2\right)^{-\frac{1}{2}}.2\theta = \frac{\theta}{|\theta|} = sign(\theta)$$

$sign(\theta) = -1$ if $\theta < 0$,
$sign(\theta) = +1$ if $\theta > 0$
$sign(\theta) = 0$ if $\theta = 0$

Applying Lasso to Linear Regression

$$L(\boldsymbol{\theta}) = \lambda\left(\sum_i (y_i - \boldsymbol{\theta}.X_i)^2\right) + (1-\lambda)\sum_j |\theta_j|$$

$$\frac{\delta}{\delta\theta_i}\sum_j |\theta_j| = sign(\theta_i)$$

$$\Delta L(\theta)$$

$$= -2\lambda\left[\sum_i X_i.(y_i - \theta.X_i)\right] + (1-\lambda)sign(\theta)$$

In matrix notation:

$$\Delta L(\theta) = -2\,\lambda\,X^T(Y - X\theta) + (1-\lambda)sign(\theta)$$

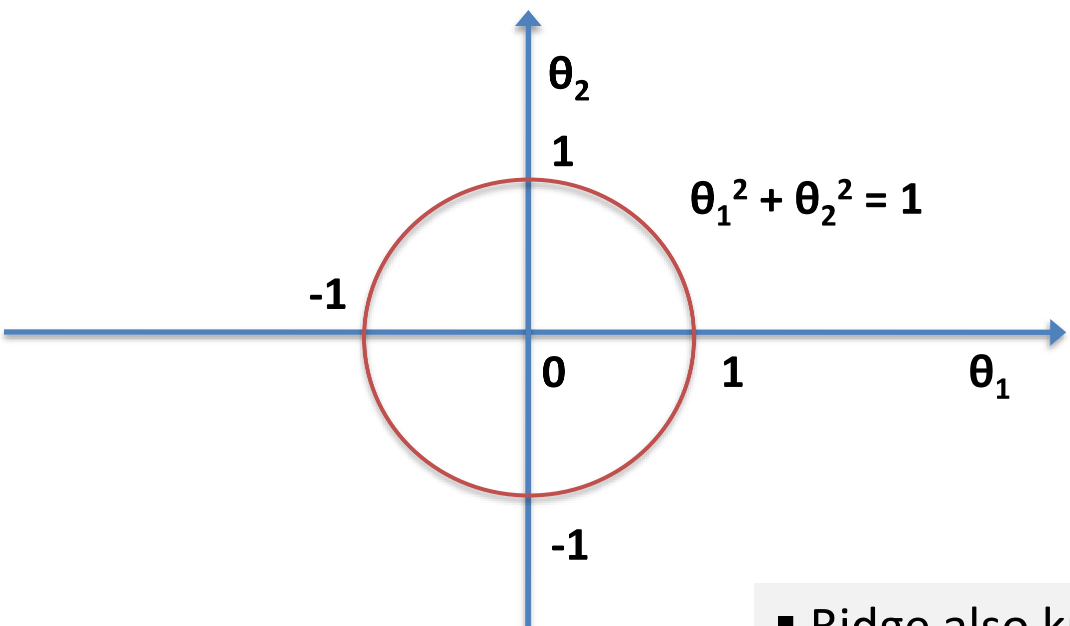
Solving Lasso

- Does not admit a Closed form solution
- Can be solved efficiently using *special purpose* algorithms
- Can be solved using *general purpose* constrained optimisation methods.
- Simple gradient based search not guaranteed to work. But we will try these nevertheless.

p = 2 : Ridge Regularized Regression

$$L(\boldsymbol{\theta}) = \lambda\left(\sum_i (y_i - \boldsymbol{\theta}.X_i)^2\right) + (1-\lambda)\sum_j |\theta_j|^2$$

- Smooth penalty over values of θ_j



Unit circle

In general:

hypersphere

- Ridge also known as L2 Regularization

Computing gradient

$$L(\theta) = \lambda\left(\sum_i (y_i - \theta . X_i)^2\right) + (1-\lambda)\sum_j |\theta_j|^2$$

$$\Delta L(\theta) = -2\lambda \sum_i X_i . (y_i - \theta . X_i) + 2(1-\lambda)\theta$$

In matrix notation:

$$\Delta L(\theta) = -2\lambda\, X^T (Y - X\theta) + 2(1-\lambda)\theta$$

Ridge Closed form solution

- Ridge admits closed form solution

$$\Delta L(\boldsymbol{\theta}) = -\mathbf{2}\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\theta}) + \mathbf{2}(\mathbf{1}-\boldsymbol{\lambda})\boldsymbol{\theta}$$

- Setting the derivative to **0**:

$$-\mathbf{2}\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}(\boldsymbol{Y} + \boldsymbol{X}\boldsymbol{\theta}) + \mathbf{2}(\mathbf{1}-\boldsymbol{\lambda})\boldsymbol{\theta} = \mathbf{0}$$

$$-\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{Y} + \boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X}\boldsymbol{\theta} + (\mathbf{1}-\boldsymbol{\lambda})\boldsymbol{\theta} = \mathbf{0}$$

$$-\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{Y} + \boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X}\boldsymbol{\theta} + (\mathbf{1}-\boldsymbol{\lambda})\boldsymbol{I}\boldsymbol{\theta} = \mathbf{0}$$

where $\boldsymbol{I}$ is the $\boldsymbol{M} \times \boldsymbol{M}$ Identity matrix (assuming $\boldsymbol{\theta}$ is $\boldsymbol{M}$-dim.)

$$-\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{Y} + [\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + (\mathbf{1}-\boldsymbol{\lambda})\boldsymbol{I}]\boldsymbol{\theta} = \mathbf{0}$$

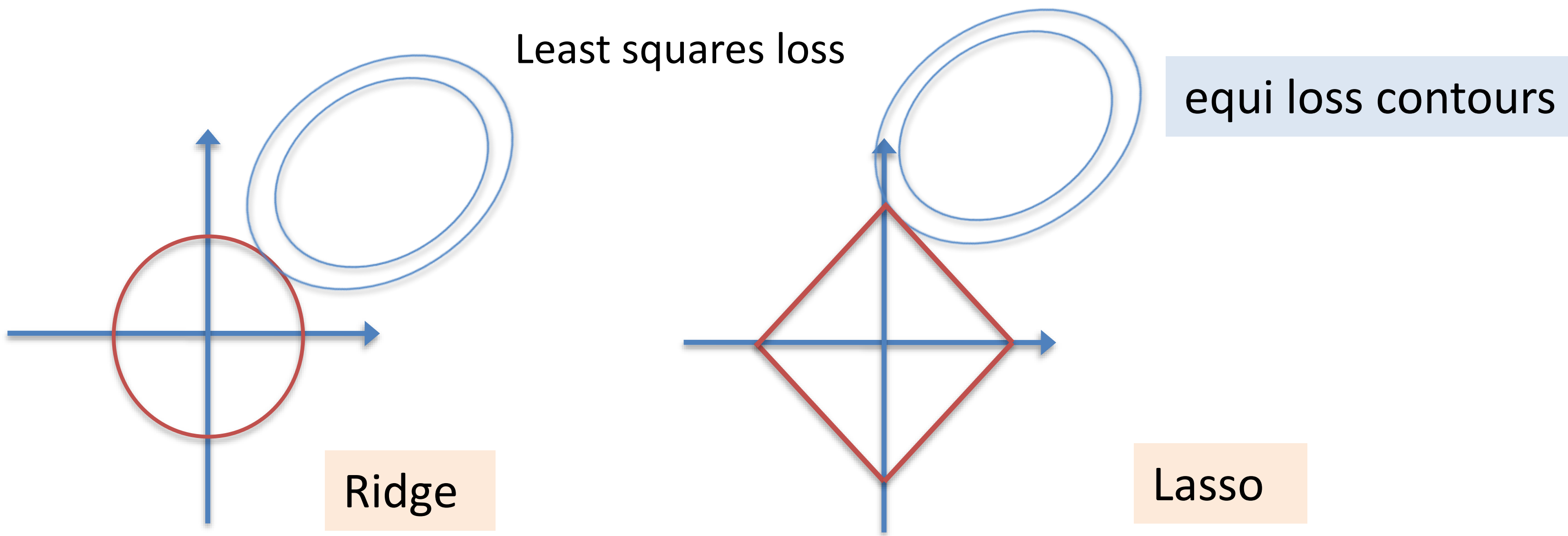
$$[\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + (\mathbf{1}-\boldsymbol{\lambda})\boldsymbol{I}]\boldsymbol{\theta} = \boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{Y}$$

$$\boldsymbol{\theta} = [\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + (\mathbf{1}-\boldsymbol{\lambda})\boldsymbol{I}]^{-\mathbf{1}}\,\boldsymbol{\lambda}\,\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{Y}$$

Solving Ridge Regression

- Can be solved using:
  - closed form solution (may be too slow)
  - gradient based methods
  - extended least squares solvers
  - constrained optimisation methods

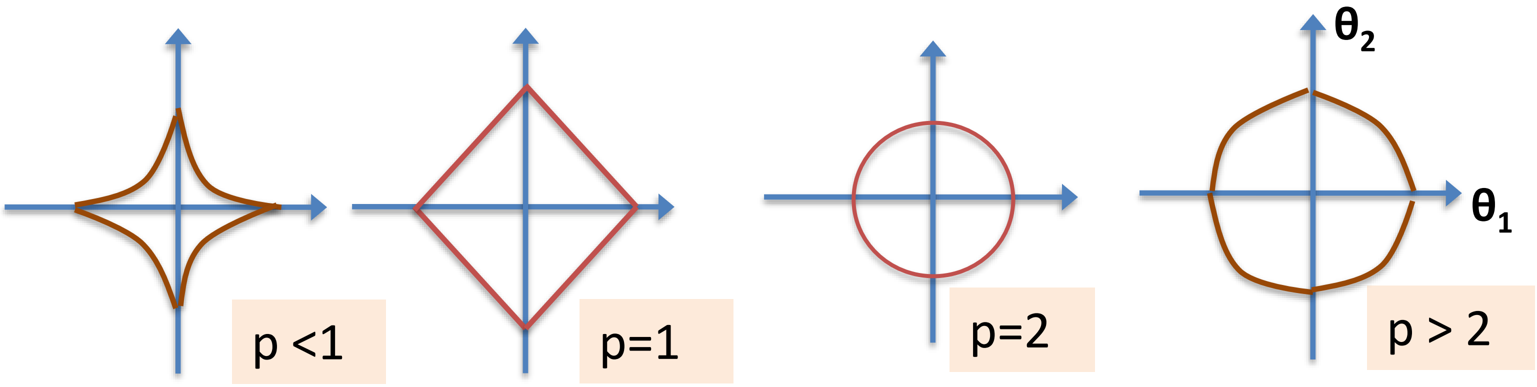
Comparison between Ridge and Lasso



- Ridge tends to shrinks all features equally
- Lasso can often set features to zero
- Hence, Lasso can be thought of as a soft version of subset selection.

Shape of $\sum|\theta_j|^p$

- two parameter case
- higher-dimensional case analogous



- function non-convex for $p < 1$
- $p = 1$ is the smallest p for which it is convex

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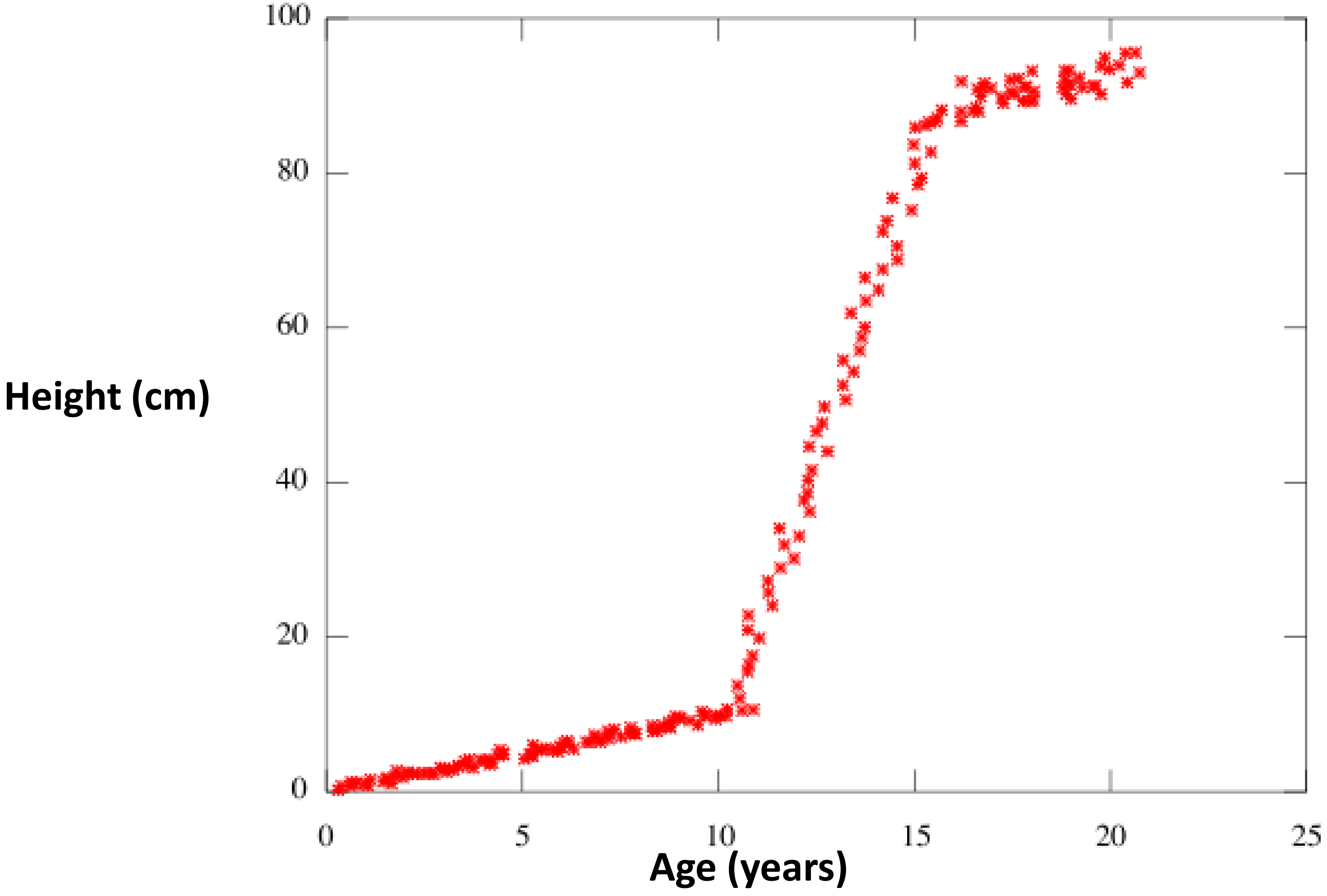
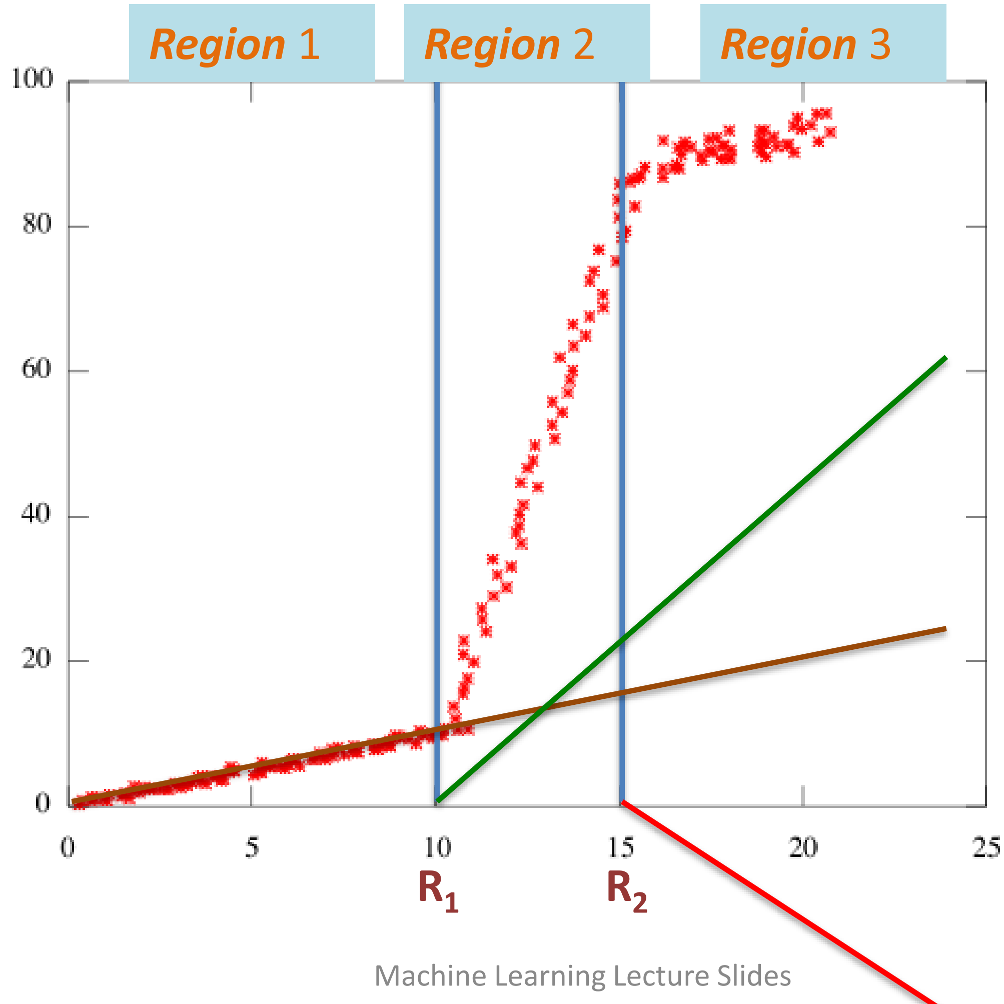
Example: Piecewise linear Age data



Remember: Fitting a piecewise polynomial

- Represent target polynomial as **sum of 3 different polynomials covering different regions**



R_1 and R_2 are called **knots**

The 3 polynomials are the 3 straight lines

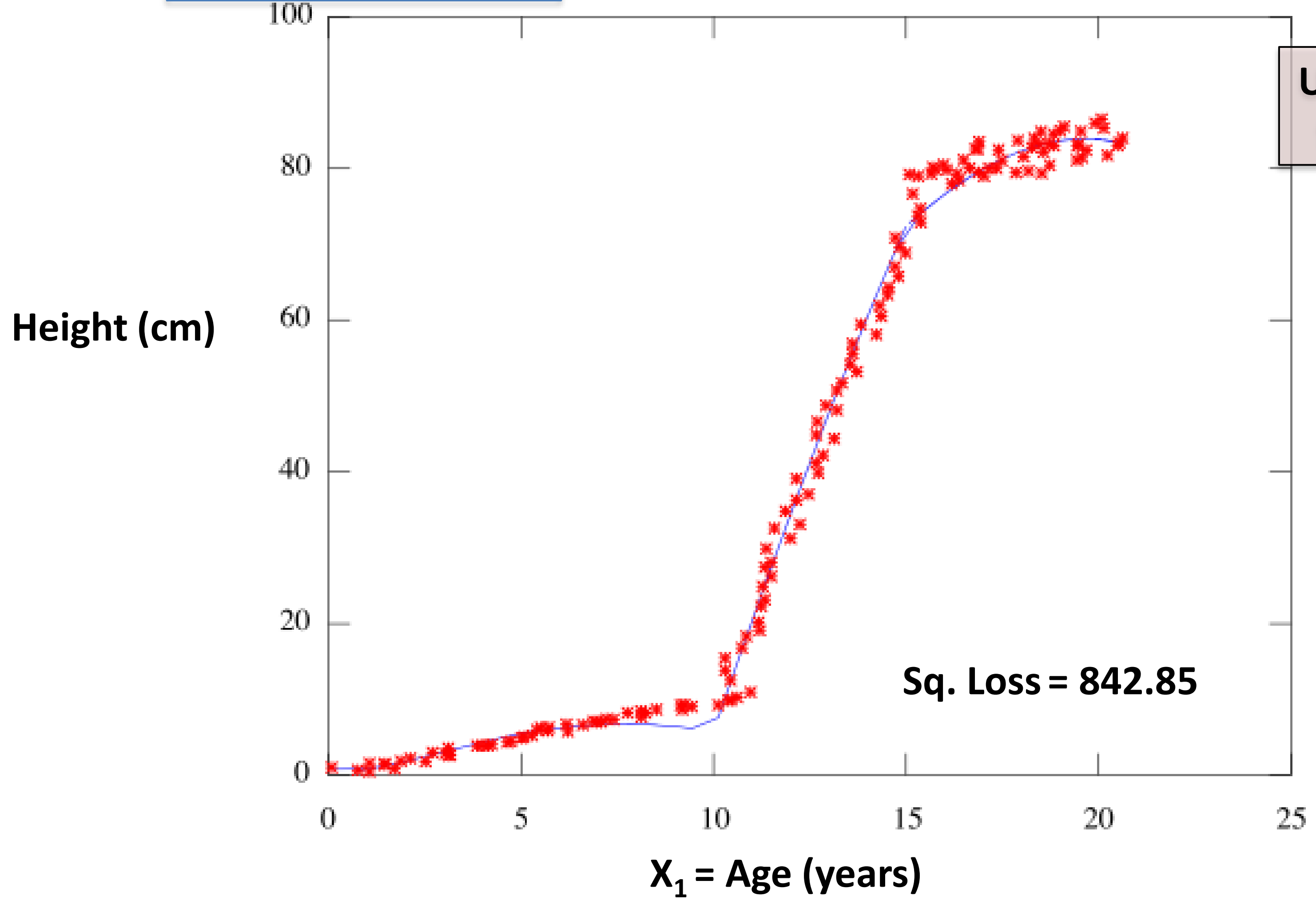
This is one possible solution.

Other alternatives are possible.

Note:
R_1 = 10
R_2 = 15

Fitting $\theta_0 + \theta_1 x_1 + \boxed{\theta_2 x_1^2 + \theta_3 x_1^3}$ $\boxed{+ \theta_4 (x_1-5)_+} + \theta_5 (x_1-10)_+ + \theta_6 (x_1-15)_+$

Unwanted features



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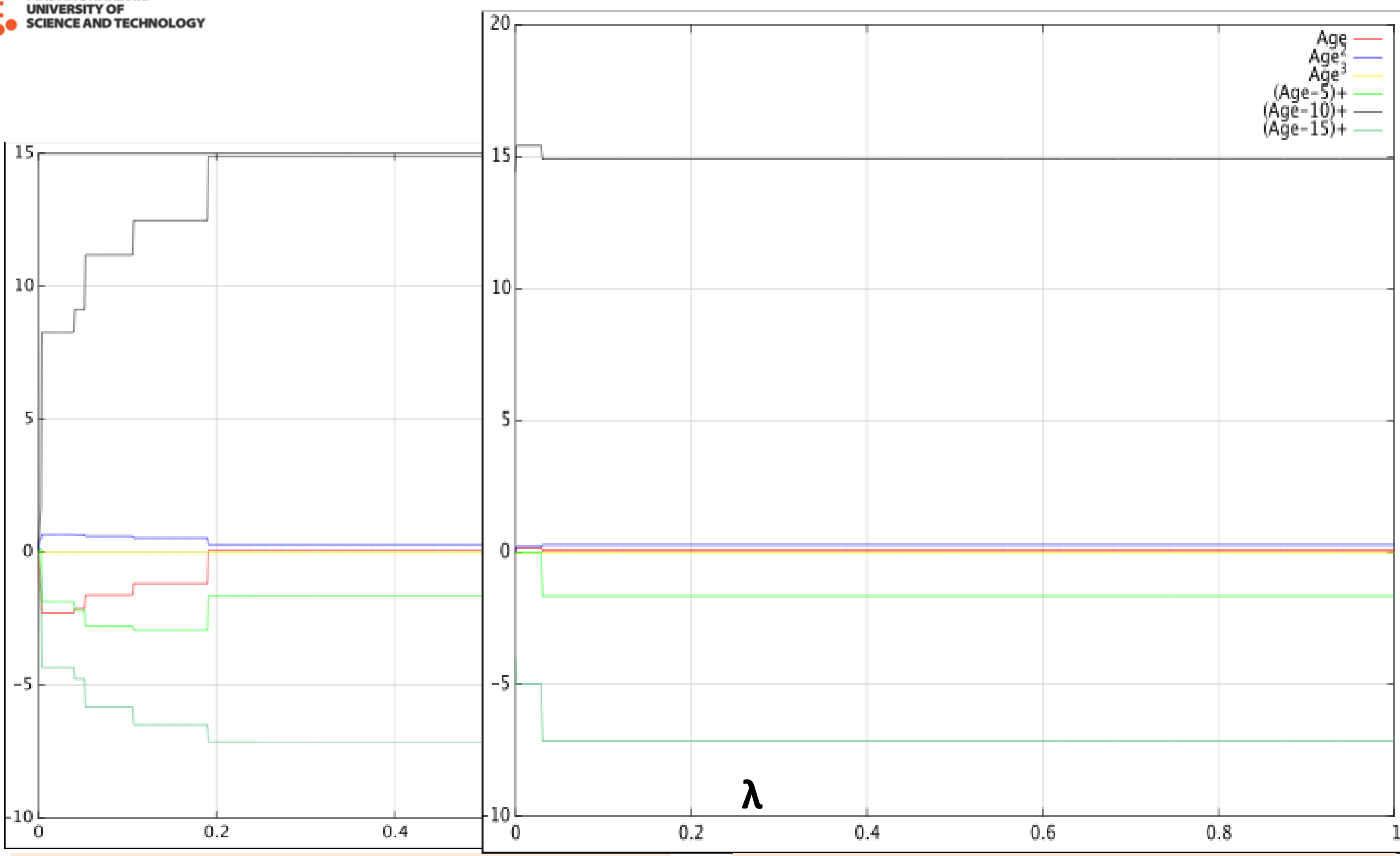
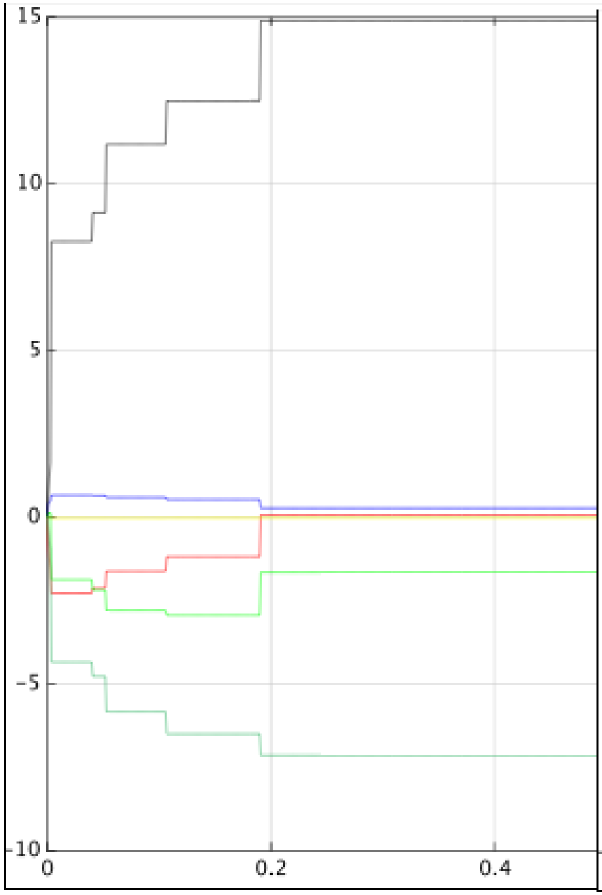


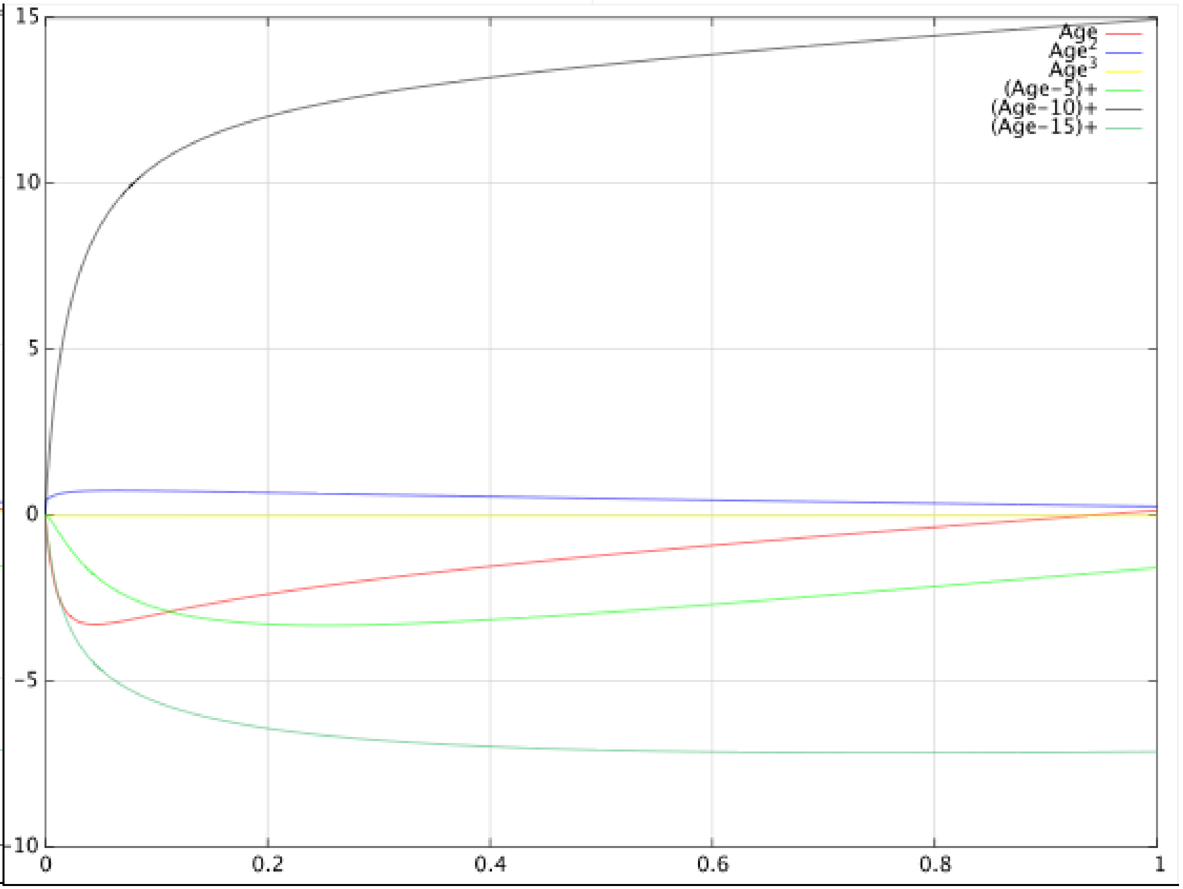
Ridge on (piecewise linear) Age data

Lasso on (piecewise linear) Age data

- For small (suitable) values of **λ** Lasso outperforms Ridge
- Models generated using fminunc so less than perfect.
- Sets **Age^2, Age^3, (Age-5)$_+$** close to 0. In particular **(Age-5)$_+$**

Ridge Gradient descent vs Closed form





Ridge using Gradient descent

Ridge using closed form solution

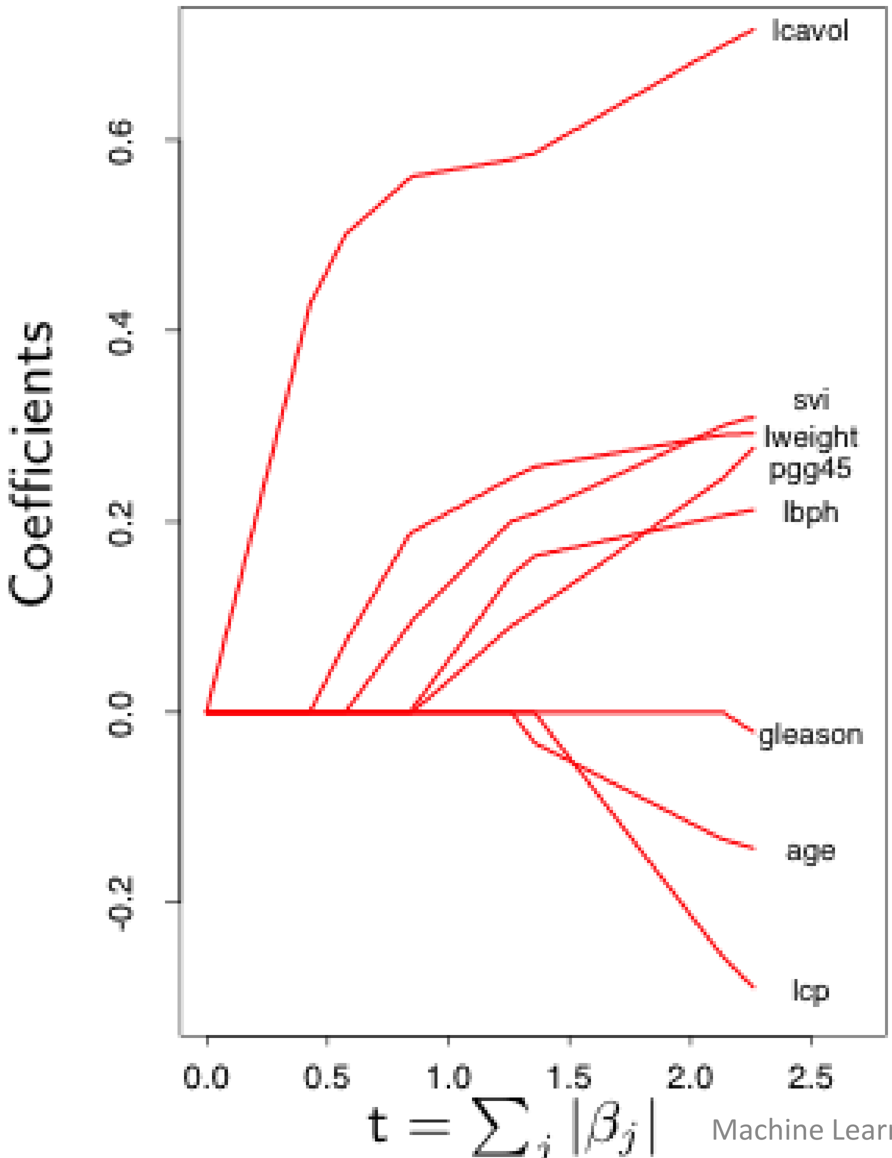
- The gradient descent solutions closely approximate the exact closed form solutions

Data preparation

- θ_0 parameter should not be included in the Regularization parameter
- We do not want to shrink this parameter
- Can either include θ_0 only in the squared loss term but remove from Regularization term.
- or mean shift Y (and X) – thus θ_0 (becomes=0) is not needed
- When ***only*** degree-1 polynomials are involved, then data scaling (i.e. scale transform) would be appropriate.

Cancer data from Stat. Learn. Book

Lasso Regression Example



- Linear **degree-1 polynomial** over all features
- **t** is the bound on the constrained optimisation problem that can be mapped to **λ**
- Many variables (lweight, pgg45, age, lcp, gleason) are set to **0** until much later as **λ** increases.
- Clearly demonstrates that Lasso is similar to soft subset selection.

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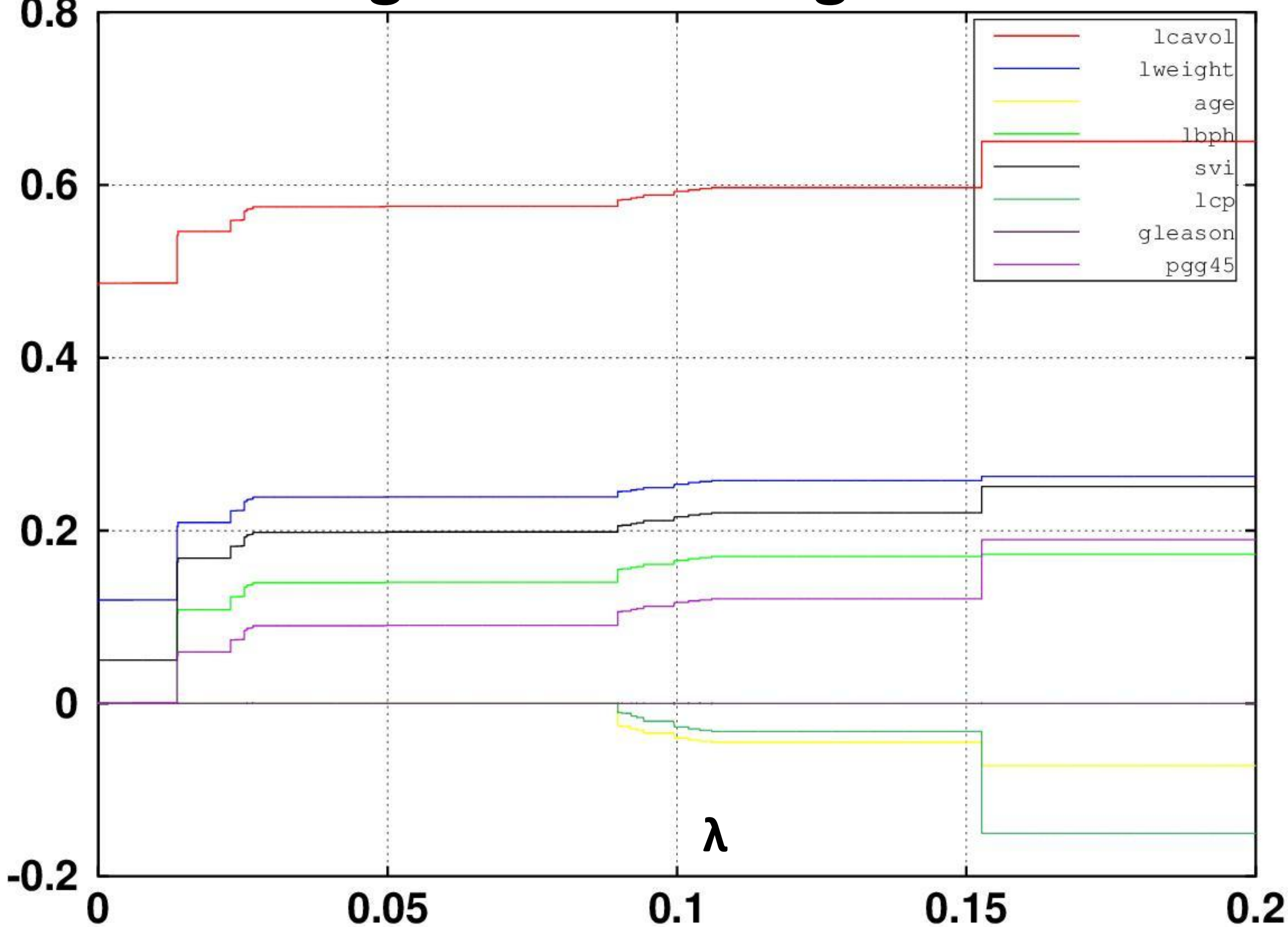
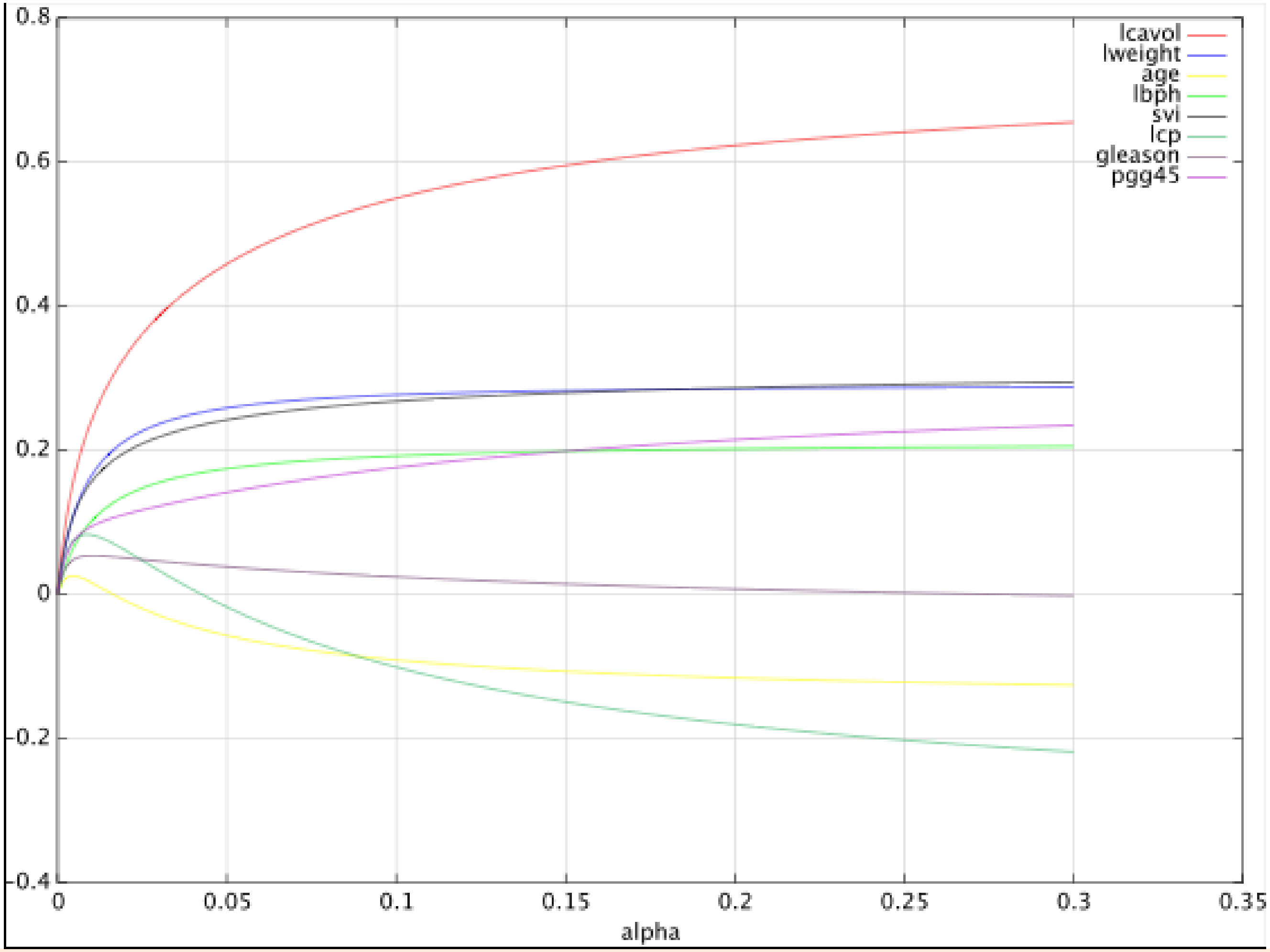
Lasso using Octave BFGS gradient descent



- Results generated using fminunc BFGS gradient descent (so not optimal)
- Very similar results (but on different scales)
- Points can be mapped from previous figure to this (roughly speaking)

Ridge using closed form solution



- The figure shows poor results using ridge regression as many variables are not shrunk to 0
- The results are quite distinct from Lasso

Homework

- Work out equations for gradient in the case of Logistic regression for Lasso and Ridge Regularization

Learning outcomes

- Understand at a high-level what a shrinkage based Regularization aims to do
- Know the general mathematical formulation
- Understand the differences between subset selection, Lasso and Ridge Regularization
- Understand how changing λ will affect the objective
- Understand why $p = 1$ is the minimum value for the regularised loss to be convex
- Be able to derived the gradient equations for linear and logistic regression using Lasso and Ridge
- Derive closed form solutions for Ridge
- Implement the regularised variants